American Computer Science League

2025-2026 ● Contest 3: Short Problems ● Senior Division

1. Boolean Algebra

How many ordered triples make the following Boolean expression FALSE?

$$A\,(\overline{B} + C) + \overline{B}(\overline{A} + C) + C(A + B)$$

A. 1
B. 2
C. 3
D. 4
E. 5

2. Boolean Algebra

Simplify the following Boolean algebra expression:

$$\overline{ABC} + \overline{\overline{B} * \overline{A + C}} + \overline{C}(A + B)$$

A. 0
B. $A\overline{C}$
C. $B\overline{C}$
D. $A\overline{C} + B\overline{C}$
E. 1

3. Data Structures

Build a binary search tree for the string

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A breadth-first traversal is listing each node starting with the root, then each row of the tree from top to bottom. Within each row, nodes are listed from left to right. What is the breadth-first traversal of this tree?

A. CCCNNOTTUEI
B. COCNCTNTEIU
C. CCOCNTNTUEI
D. CCCEINNOTTU
E. COCTNCUTNEI

4. Data Structures

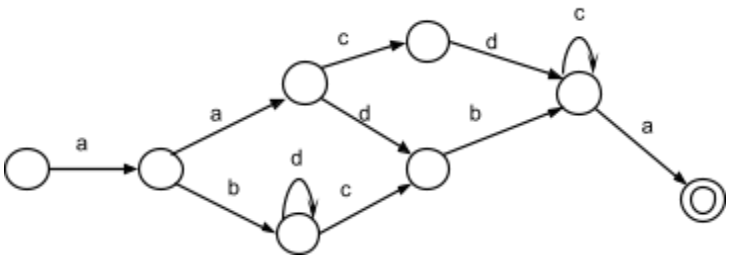
Which of the following strings, when used in the following algorithm on an initially empty stack, will result in a stack that contains the original string?

```
PUSH(first letter in string)
for each other letter in the string
        x = POP()
        PUSH(x)
        if letter != x then
                PUSH(letter)
        end if
next letter
```

A. APPLES
B. CARROT
C. PEPPER
D. ORANGE
E. CHERRY

5. FSAs

Which letter is required to be in all of the accepted strings?



A. a
B. b
C. c
D. d
E. More than 1 of them

6. Regular Expressions

In ACSLland license plates are issued using the following regular expression:

```
[D-R1-5]* [^0-2][a-t]?
```

How many of the following license plates are accepted?

J44ee	EN811f	IR54p	FO795m
PH94n	GM223g	Dan	48ACSL

A. all of them
B. 7
C. half of them
D. 1
E. none of them